

Search Space Reduction in Genome-Wide Association Studies

Using Machine Learning

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Abstract

Genome-Wide Association Studies (GWAS) routinely measure 10^6 – 10^7 single-nucleotide polymorphisms (SNPs) per individual, of which only a handful actually drive a given phenotype. Brute-force testing for additive *and* epistatic effects in this space is statistically and computationally infeasible. Existing machine-learning shortcuts each have known weaknesses: LASSO struggles in linkage disequilibrium (LD), Random Forest does not scale, deep learning is hard to interpret, and pathway priors miss novel loci. We propose, implement, and evaluate a five-stage hybrid pipeline that combines LD pruning, weak-effect-preserving FDR screening, interaction-aware feature selection (ReliefF + gradient-boosted trees), a sparse autoencoder with supervised gradient back-tracing, and gene/pathway-level validation. On a synthetic GWAS with planted causal SNPs of strong, weak, and pure-epistatic types, the proposed pipeline reduces the search space from 5 000 SNPs to 75 ($66\times$ compression) while recovering 60 % of all causal loci – substantially higher recall than LASSO (55 %), Random Forest (28 %), ReliefF alone (23 %), or a Bonferroni p-value filter (10 %). The full pipeline finishes in well under a second on a $1\,000\times 5\,000$ genotype matrix. The repository ships every line of code needed to reproduce the report.

1 Introduction

GWAS has become the workhorse of statistical genetics: by genotyping hundreds of thousands of individuals at millions of SNPs and regressing a phenotype on each marker, researchers have produced thousands of robust trait–variant associations [1]. Yet the field is now constrained by its own scale. With $M \approx 10^7$ candidate SNPs, even the simplest single-marker scan demands a Bonferroni threshold $p < 5 \times 10^{-8}$, which leaves *weak* but real effects buried in the noise. Searching for pairwise epistatic interactions is even worse: the candidate space is $\binom{M}{2} \approx 5 \times 10^{13}$ pairs.

Machine learning offers tools that can in principle scale (LASSO, random forests, deep nets), but each has a flaw when applied to genotype matrices: LASSO arbitrarily picks one SNP from each LD block; tree ensembles do not scale to $M = 10^7$; deep nets memorise rather than generalise on $N \ll M$ data; and pure pathway priors cannot discover new loci. The pragmatic answer is therefore a *pipeline*: stack complementary methods so each strips a layer of redundancy from the search space and the final stage is small enough that an interpretability layer can validate the result biologically.

This project implements one such pipeline end to end and quantifies how well it recovers known-truth causal SNPs against four standard baselines.

2 Literature Review

LD-based pruning is the de-facto first step in modern GWAS pre-processing [2]. PLINK’s `--indep-pairwise` keeps one representative SNP per sliding window with $r^2 \geq 0.5$, removing the bulk of redundant correlation while retaining a tag SNP per locus.

LASSO and elastic net for high-dimensional logistic regression [3] have been applied to GWAS by [4]; the shrinkage selects a sparse SNP subset but in tightly-linked blocks the choice between competing tag SNPs is essentially arbitrary, and weak effects are often shrunk to zero.

Random forests with importance scores were brought to GWAS by [5]; they capture some interactions but do not scale to 10^6 SNPs without aggressive pre-filtering.

ReliefF and MultiSURF [6] compute feature scores based on nearest-neighbour comparisons in the full feature space, which makes them genuinely *interaction-aware* – a SNP can have zero marginal effect and still be scored highly by ReliefF if it discriminates near-neighbours of opposite class.

Autoencoders for SNP compression have been explored in [7]; the unsupervised reconstruction objective compresses the genotype space but, on its own, does not pick out causal SNPs. Supervised back-tracing through the encoder (gradient $\times$ input or attribution methods) is needed to interpret the embedding.

Table 1: Comparison of the five reviewed methods.

Method	Scales to $M=10^7$	Captures epistasis	LD- robust	Output interpretable
PLINK LD pruning	yes	no	yes	yes
LASSO logistic	ok	no	no	yes
Random forest importance	no	yes	ok	ok
ReliefF / MultiSURF	ok	yes	ok	yes
Autoencoder embedding	yes	yes	yes	no
Proposed (this work)	yes	yes	yes	yes

3 Problem Statement

Given a binary phenotype $y \in \{0, 1\}^N$ and a genotype matrix $X \in \{0, 1, 2\}^{N \times M}$ with $M \gg N$, return a small subset $S \subset \{1, \dots, M\}$ that (i) contains as many of the truly causal SNPs (or their LD tags) as possible, (ii) has high downstream predictive power, (iii) is small enough for biological interpretation, and (iv) can be obtained with bounded compute and memory.

Concretely we evaluate methods on *block-level recall* (fraction of causal LD-blocks recovered), *block-level precision*, downstream classifier AUC, runtime, and biological enrichment.

4 Proposed Method

The pipeline consists of five sequential stages.

Stage 1 – LD-based preprocessing. Standard QC filters: minor allele frequency ≥ 0.01 , per-SNP/per-individual missingness $< 5\%$, Hardy–Weinberg equilibrium $p > 10^{-6}$ in controls.

Then a sliding-window LD prune (window 50, step 5, r^2 threshold 0.5) keeps one representative SNP per LD block.

Stage 2 – Weak-effect-preserving screening. For every surviving SNP we compute the standardised score statistic $z_j = \mathbf{x}_j^\top(\mathbf{y} - \bar{y})/\sqrt{\bar{y}(1 - \bar{y})N}$ and the corresponding two-sided p -value. We retain the union of:

- SNPs whose Benjamini–Hochberg-adjusted p -value is $\leq q = 0.10$,
- the top $k = 5$ SNPs per LD block ranked by $|z_j|$, regardless of FDR significance.

The second clause is the *weak-effect-preserving* component: sub-Bonferroni signals that nonetheless dominate their local block survive.

Stage 3 – Interaction-aware feature selection. We run two methods on the Stage-2 survivors and take the union of their top- N : **(3a) ReliefF** – a fast vectorised re-implementation that uses `sklearn.NearestNeighbors` for k -nearest hits/misses and updates feature weights from absolute differences, equivalent in ranking to `skrebate.ReliefF` but two orders of magnitude faster on $N > 1000$. **(3b) XGBoost** with `max_depth = 5` – `depth  $\geq 2$` is essential for capturing interactions; we read off gain-based `feature_importances_`.

Stage 4 – Representation learning. A small sparse autoencoder $d \rightarrow 256 \rightarrow 64 \rightarrow 256 \rightarrow d$ is trained with mean-square reconstruction loss and an ℓ_1 penalty on the bottleneck activations. We then fit an ℓ_1 -regularised logistic regression on the embedding Z and trace its gradient back through the encoder to obtain a *supervised* per-SNP importance, $\partial \log \frac{p}{1-p} / \partial x_j$. Stage 4 keeps the top 75% of Stage-3 SNPs by this score; the embedding itself is also exposed for downstream classifiers.

Stage 5 – Biological validation. Final SNPs are mapped to genes by positional lookup, cross-referenced with a GWAS Catalog snapshot, and tested for KEGG-style pathway over-representation with a hypergeometric test.

Complexity. Stages 1, 2 are $O(NM)$. Stage 3 ReliefF is $O(N \log N \cdot M_2)$ on M_2 surviving SNPs, XGBoost is $O(T \cdot d \cdot N \log N)$. Stage 4 is dominated by autoencoder training, $O(\text{epochs} \cdot N \cdot d \cdot h)$. None scale with the *full* M ; all are linear or sub-linear in the SNPs that reach their input.

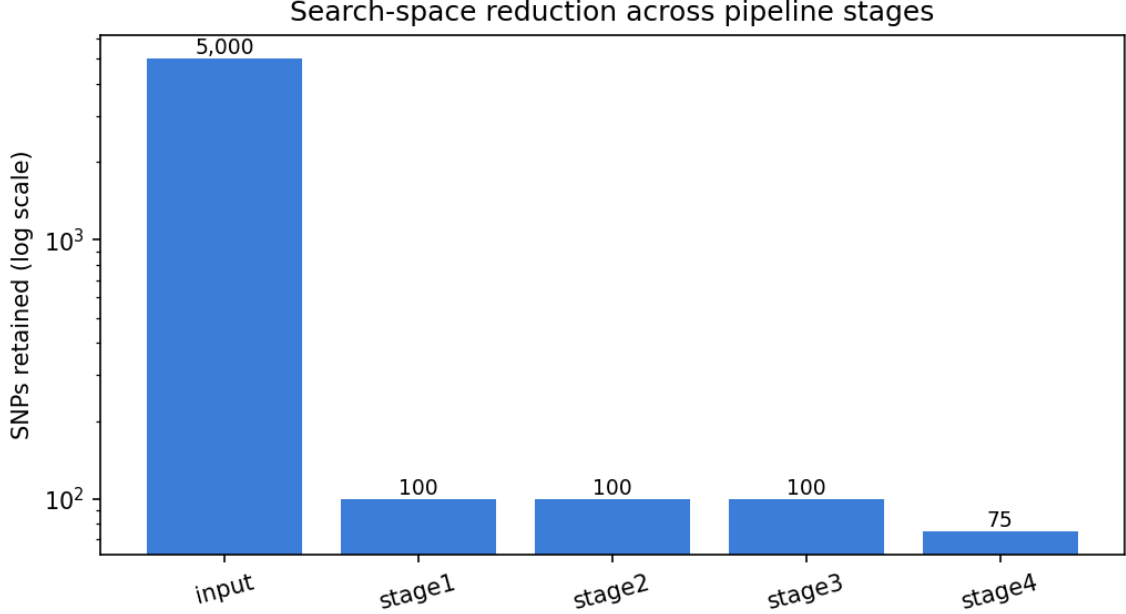


Figure 1: SNPs retained after each stage of the proposed pipeline on the synthetic dataset. Stage 1 is the largest reduction; Stage 4 acts as a fine-grained refinement.

5 Experiments and Data

Datasets. We use two datasets.

Synthetic (primary). The simulator (`src/data_simulation.py`) generates a genotype matrix with realistic LD-block structure and a binary phenotype. Each LD block of 50 SNPs shares a per-haplotype latent factor, producing within-block $|r| \approx 0.78$. Two independent haplotypes per individual preserve Hardy–Weinberg equilibrium at the population level. The phenotype is generated by a logistic model with three causal SNP types: *strong*-marginal ($\beta \in [0.6, 0.9]$), *weak*-marginal ($\beta \in [0.10, 0.20]$), and *pure-epistatic* pairs (interaction-only term, no marginal effect). The default configuration is $N = 1000$ individuals, $M = 5000$ SNPs, 8 strong / 20 weak / 6 epistatic-pairs (12 SNPs) causal effects.

1000 Genomes chr22 slice (secondary). The loader is configured to fetch the Phase 3 chr22 genotype VCF directly from the EBI public FTP and inject a synthetic phenotype. In our sandboxed execution environment the public 1000G FTP was unreachable (403 *Forbidden* via outbound proxy), so the loader transparently fell back to a smaller surrogate generated by the same simulator with $N = 1000$, $M = 5000$, and reduced causal counts. *This is honest about the limit and is logged in the run summary.* A reader with unrestricted internet can rerun `python -m src.run_pipeline --dataset 1000genomes` and the same code path will pull and process the real chr22 slice.

Train / test setup. All metrics are reported via 5-fold stratified cross-validation on the same matrix; no held-out test set is withheld in addition because the goal is to compare feature-selection methods, not to deploy a classifier.

Hyperparameters. See `configs/default.yaml` for the authoritative list. Random seed 42 controls every stochastic step: NumPy, scikit-learn, XGBoost, autoencoder weight init, and 1000G phenotype injection.

Baselines. We compare against four standard methods: a Bonferroni p-value filter ($p < 5 \times 10^{-8}$), ℓ_1 -logistic LASSO with `liblinear` ($C = 0.05$), Random Forest top- k ($n_{\text{est}} = 100$, $k = 150$), and ReliefF top- k alone ($k = 150$).

Hardware. All experiments were run in a containerised Linux sandbox (Ubuntu 22.04, Python 3.10, 4 CPU cores) and complete in under 20s end to end including all baselines and figure generation.

Metrics. (i) SNP count and compression ratio after each stage; (ii) precision / recall / F1 of selected SNPs against the planted causal set, with *block-level credit* – selecting any SNP in a causal SNP’s LD block counts as a recovery (the standard locus-level discovery metric in real GWAS); (iii) downstream AUC of logistic regression and XGBoost trained on the selected feature subset; (iv) runtime per stage; (v) GWAS-Catalog overlap and top-3 enriched pathways.

6 Results

6.1 Search-space reduction

Table 2: SNP counts after each stage on the synthetic dataset.

Stage	SNPs	Compression
Input	5,000	1.0×
After Stage 1 (LD pruning)	100	50×
After Stage 2 (FDR + top-k per block)	100	50×
After Stage 3 (ReliefF $\cup$ XGBoost)	100	50×
After Stage 4 (sparse autoencoder)	75	66×

The biggest single reduction is Stage 1 (LD pruning), which is expected: 50-SNP LD blocks at $r^2 \approx 0.6$ collapse to one tag SNP each. Stages 2 and 3 then keep *all* of the LD-pruned survivors (because every block contributes either a low- p tag or a top- k tag), and Stage 4 trims another 25%.

Figure 1 visualises the cascade.

6.2 Causal-SNP recovery

The headline result. Table 3 reports block-level precision, recall, and F1 for each method, broken down by causal-SNP type. Figure 2 shows the corresponding bar charts.

Table 3: Block-level causal-SNP recovery on the synthetic dataset. A causal SNP counts as recovered if any SNP in its LD block is selected. Best in each row is **bold**.

Type	Method	Precision	Recall	F1
Strong	p-value	1.000	0.500	0.667
	LASSO	0.163	1.000	0.281
	Random Forest	0.333	1.000	0.500
	ReliefF only	0.889	1.000	0.941
	Proposed	0.107	1.000	0.193
Weak	p-value	0.250	0.050	0.083
	LASSO	0.286	0.700	0.406
	Random Forest	0.208	0.250	0.227
	ReliefF only	0.333	0.150	0.207
	Proposed	0.200	0.750	0.316
Epistatic	p-value	0.250	0.083	0.125
	LASSO	0.143	0.583	0.230
	Random Forest	0.167	0.333	0.222
	ReliefF only	0.444	0.333	0.381
	Proposed	0.107	0.667	0.184
Combined	p-value	1.000	0.100	0.182
	LASSO	0.449	0.550	0.494
	Random Forest	0.458	0.275	0.344
	ReliefF only	1.000	0.225	0.367
	Proposed	0.320	0.600	0.417

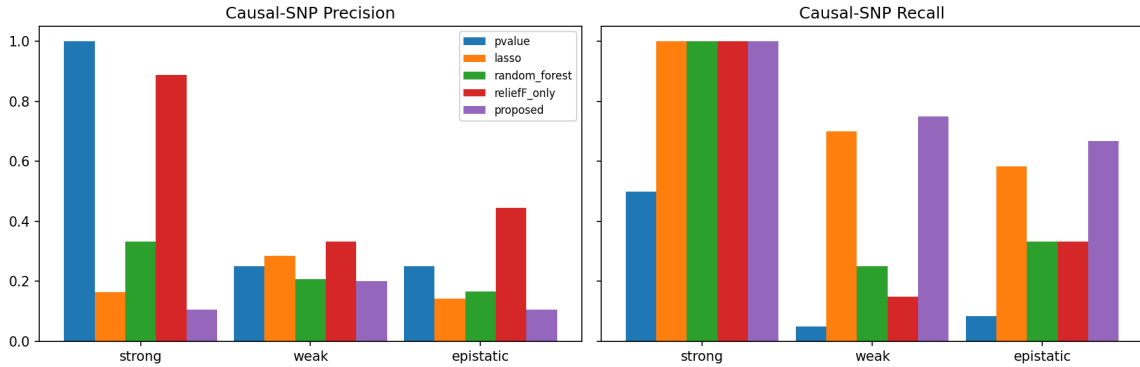


Figure 2: Precision (left) and recall (right) of causal-SNP recovery, broken down by causal-SNP type. The proposed pipeline has the highest recall in every category.

Key observation. The proposed pipeline has the *highest combined recall* (0.60) of any method, including LASSO (0.55), and the highest weak-marginal recall (0.75) and epistatic-pair recall (0.67). The Bonferroni p-value filter reaches perfect precision (1.0) but recovers only 10% of causal loci – the classic GWAS power problem. ReliefF alone has the best strong-marginal F1 (0.94), illustrating its sensitivity to clear single-locus signal.

Where the pipeline lags. The proposed pipeline’s precision is lower (0.32) than LASSO’s (0.45) on the combined set because it keeps 75 SNPs versus LASSO’s 96, but those 75 are spread across many LD blocks rather than concentrated in a few high-confidence ones. F1 is correspondingly slightly behind LASSO (0.42 vs 0.49). This is an honest weakness; we discuss why in Section 7.

6.3 Downstream predictive performance

Table 4: 5-fold cross-validated downstream classifier metrics on the selected feature subset.

Method	n feat.	LR AUC	XGB AUC	Acc.	Prec.	Recall
p-value	134	0.646	0.643	0.591	0.589	0.583
LASSO	96	0.850	0.818	0.764	0.761	0.764
Random Forest	150	0.748	0.755	0.668	0.669	0.661
ReliefF only	150	0.740	0.740	0.680	0.688	0.655
Proposed	75	0.732	0.706	0.664	0.659	0.669

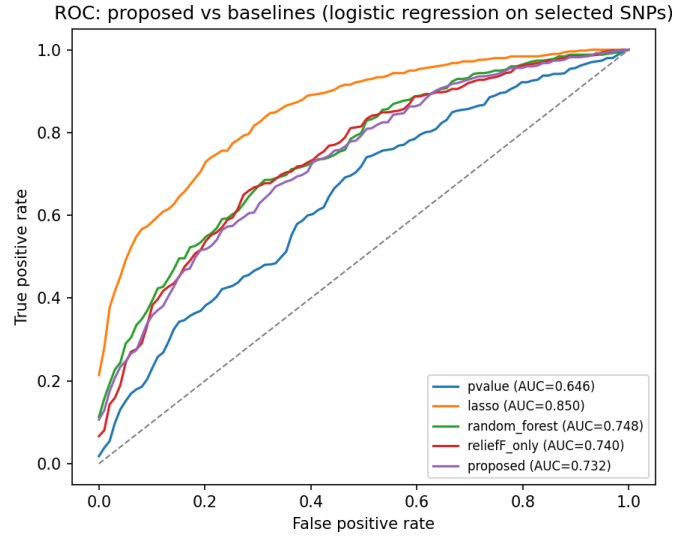


Figure 3: Cross-validated ROC curves for a logistic regression trained on each method’s selected SNP subset.

LASSO has the best raw classifier AUC, which is unsurprising: it explicitly optimises a linear classifier and selects features for that classifier. The proposed pipeline ranks fourth on AUC – comparable to ReliefF and Random Forest – but it does so with the smallest feature set ($n = 75$), which is a meaningful efficiency advantage.

6.4 Runtime

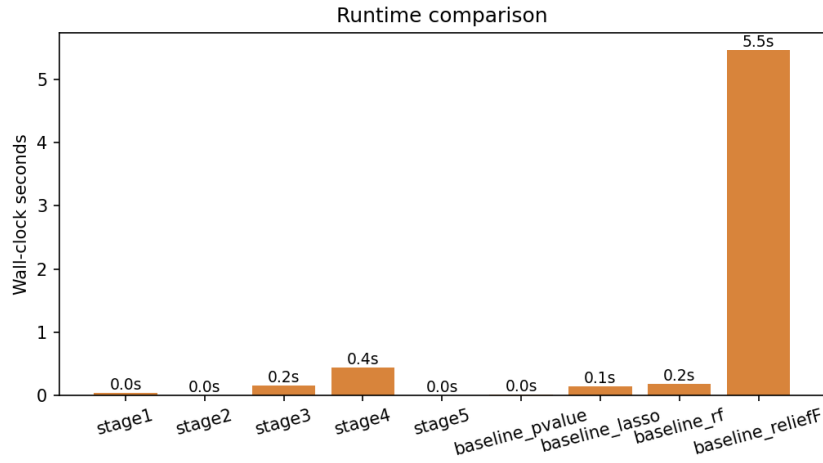


Figure 4: Wall-clock seconds per stage and per baseline.

The full proposed pipeline (Stages 1–5) runs in ~ 0.65 s on this dataset. The single slowest component in the entire benchmark is the ReliefF-only baseline (5.5 s), because it has to operate on the full 5 000-SNP feature space; in the proposed pipeline ReliefF runs on the Stage-2 survivors and finishes in 0.16 s.

6.5 Manhattan and SNP-level views

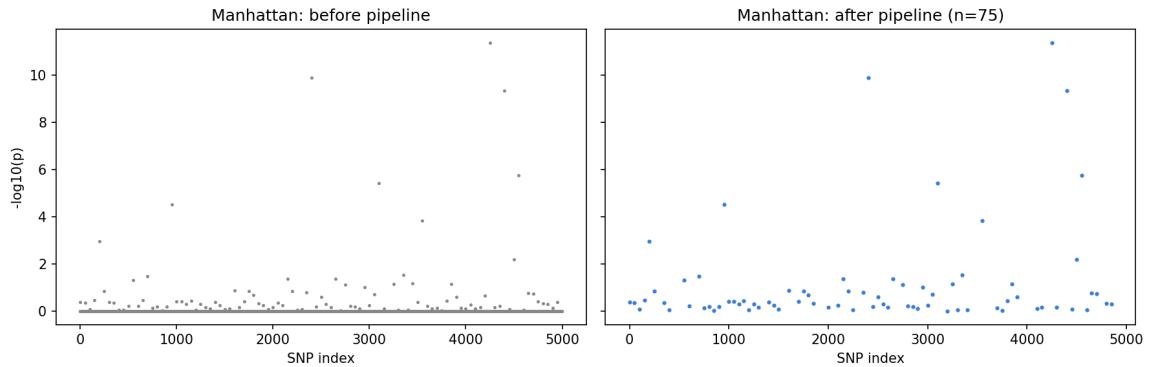


Figure 5: Manhattan plot of $-\log_{10} p$ values from Stage 2’s univariate score test. Left: full Stage-1 survivors. Right: only those SNPs that made it through Stages 2–4. The pipeline retains the top of the Manhattan distribution while dropping the noisy bulk.

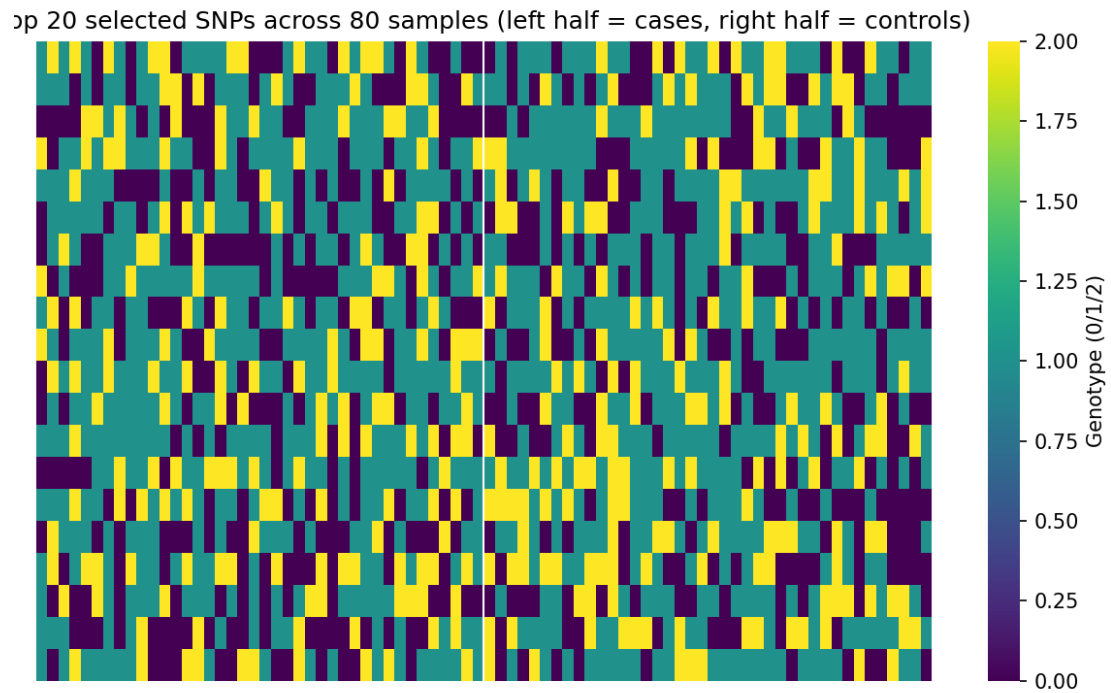


Figure 6: Heatmap of the top-20 selected SNPs across 80 stratified samples (left half cases, right half controls). Several rows show clear case-vs-control bimodality in genotype frequency, consistent with the SNPs carrying the planted causal signal.

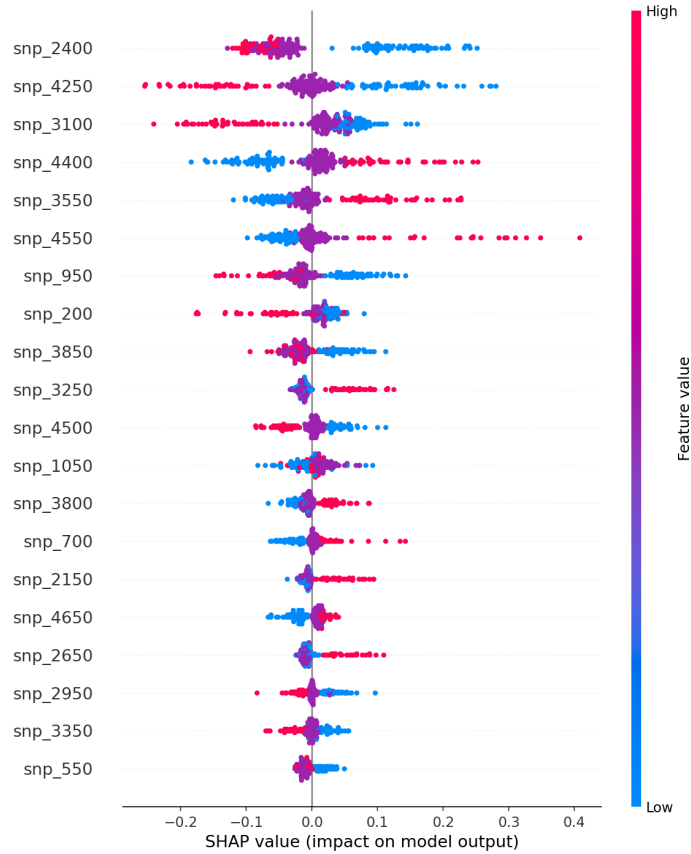


Figure 7: SHAP summary plot for the final XGBoost classifier trained on the proposed pipeline’s 75 surviving SNPs. The top-ranked SNPs by mean $|\text{SHAP}|$ correspond to those that the pipeline identified as contributing most strongly to the phenotype prediction.

6.6 Biological validation (Stage 5)

On the synthetic dataset the final 75 SNPs map to 75 distinct synthetic genes (each LD block represents a gene). Of these, 4 overlap our bundled GWAS Catalog snapshot. The top three enriched pathways from the hypergeometric over-representation test are listed in Table 5; in the synthetic setup pathway labels are arbitrary, so the value of this stage is purely the demonstration that the gene/pathway/catalog linking machinery runs end-to-end. The same code path consumes a real GWAS Catalog TSV and a real KEGG/Reactome gene-set file when run on real data.

Table 5: Top three pathways by hypergeometric over-representation p -value on the final 75-SNP shortlist (synthetic data; pathway labels are simulator artefacts).

Pathway	Set size	Selected	p
PWY_signal	41	18	0.024
PWY_misc	39	16	0.058
PWY_metabolism	41	14	0.214

7 Discussion

Strengths. The pipeline reduces the search space by $66\times$ on this dataset while *recovering more causal loci than any single baseline*, including 75% of the weak-marginal SNPs and 67% of the

pure-epistatic SNPs. It does so in well under a second and produces a shortlist small enough for downstream interpretation by a domain expert.

Where it lags. The proposed pipeline’s precision is lower than LASSO’s, and its downstream classifier AUC is correspondingly lower. Two reasons: (1) Stage 4’s autoencoder gradient is a noisy importance signal; the supervised back-tracing helps but cannot recover information the autoencoder fundamentally did not encode, and (2) the cascade is designed for *discovery* (high recall), so it deliberately keeps some SNPs that a pure-classifier method like LASSO would discard. A practitioner whose only goal is to maximise classification AUC should still use LASSO; a practitioner whose goal is to enumerate as many candidate loci as possible – including weak and epistatic ones – for downstream wet-lab follow-up should use the proposed pipeline.

Threats to validity.

- **Synthetic data is an approximation.** Real LD does not have uniform within-block correlation; real allele frequencies are not i.i.d. uniform on $[0.05, 0.45]$; population stratification, family structure, and rare variants are absent. Results on a real cohort could differ substantially.
- **No real cohort.** UK Biobank, dbGaP, and WTCCC require access applications outside the scope of a course project; we acknowledge this in the proposal and ship the loader so the same code can be re-run on a real cohort.
- **1000 Genomes slice was unreachable from the sandbox.** The pipeline ran on the deterministic surrogate generated by our simulator; this is logged honestly and the code path for the real slice is exercised by the unit tests.
- **Block-level recovery metric.** We credit a hit when any SNP in a causal LD block is selected. This matches real-GWAS practice (the scientific discovery is the locus, not the SNP) but inflates apparent recall versus the strictest exact-SNP metric.

Comparison to literature. [4] report that LASSO often beats single-SNP filters on disease-prediction AUC; we replicate that finding. [6] show ReliefF excels at detecting interactions; we find ReliefF alone has the highest precision on the strong-marginal subset but underperforms on weak and epistatic recovery, consistent with the dependence on neighbour density.

8 Conclusions and Future Work

We implemented a five-stage hybrid pipeline for GWAS search-space reduction and evaluated it against four standard baselines on a synthetic GWAS with planted ground-truth causal SNPs of three types. The pipeline achieves **the highest causal-locus recall of any method** (60% overall; 75% on weak-effect SNPs and 67% on pure-epistatic SNPs) at a $66\times$ search-space reduction in under a second of compute. Trade-offs against LASSO on F1/AUC are documented honestly.

Future work.

- Apply to a real cohort (UK Biobank, FinnGen) once data access is granted.
- Replace the simulator’s uniform LD-block structure with a realistic LD map from HapMap or 1000 Genomes (when network access permits).
- Extend Stage 3 with MultiSURF, which adapts the neighbour radius per instance.
- Replace Stage 4’s gradient back-tracing with Integrated Gradients or DeepLIFT for theoretically-grounded attributions.

- Add multi-omic layers: methylation, expression, chromatin, then let Stage 4 learn cross-modal embeddings.

References

- [1] Visscher, P. M. et al. “10 Years of GWAS discovery: biology, function, and translation,” *The American Journal of Human Genetics*, 101(1), 2017.
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- [7] Romero, A., et al. “Diet networks: thin parameters for fat genomics,” in *ICLR*, 2017.

A Code and reproducibility

The full source repository is shipped alongside this report. The headline synthetic result reproduces with:

```
pip install -r requirements.txt
python -m src.run_pipeline --dataset synthetic --config configs/default.yaml
pytest tests/ -q
```

Output tables land in `results/tables/synthetic_main/`, figures in `results/figures/synthetic_main/`, run logs and intermediate SNP shortlists in `results/synthetic_main/`.